Rewarding Learning

ADVANCED
General Certificate of Education
2026

Digital Technology

Assessment Unit A2 1

assessing

Information Systems

[ADT11]

FRIDAY 5 JUNE, MORNING

MARK SCHEME

General Marking Instructions

Introduction

Mark schemes are published to assist teachers and students in their preparation for examinations. Through the mark schemes teachers and students will be able to see what examiners are looking for in response to questions and exactly where the marks have been awarded. The publishing of the mark schemes may help to show that examiners are not concerned about finding out what a student does not know but rather with rewarding students for what they do know.

The Purpose of Mark Schemes

Examination papers are set and revised by teams of examiners and revisers appointed by the Council. The teams of examiners and revisers include experienced teachers who are familiar with the level and standards expected of students in schools and colleges.

The job of the examiners is to set the questions and the mark schemes; and the job of the revisers is to review the questions and mark schemes commenting on a large range of issues about which they must be satisfied before the question papers and mark schemes are finalised.

The questions and the mark schemes are developed in association with each other so that the issues of differentiation and positive achievement can be addressed right from the start. Mark schemes, therefore, are regarded as part of an integral process which begins with the setting of questions and ends with the marking of the examination.

The main purpose of the mark scheme is to provide a uniform basis for the marking process so that all the markers are following exactly the same instructions and making the same judgements in so far as this is possible. Before marking begins a standardising meeting is held where all the markers are briefed using the mark scheme and samples of the students' work in the form of scripts. Consideration is also given at this stage to any comments on the operational papers received from teachers and their organisations. During this meeting, and up to and including the end of the marking, there is provision for amendments to be made to the mark scheme. What is published represents this final form of the mark scheme.

It is important to recognise that in some cases there may well be other correct responses which are equally acceptable to those published: the mark scheme can only cover those responses which emerged in the examination. There may also be instances where certain judgements may have to be left to the experience of the examiner, for example, where there is no absolute correct response – all teachers will be familiar with making such judgements.

Listing

Where **one** response is required to gain a mark, candidates will not gain credit if a correct response is given alongside one or more incorrect responses. This is referred to as listing.

AVAILABLE MARKS

1 (a) **IP address**
A unique address
… identifying a device on a network
… allowing information to be sent on the network
(3 × [1])

Wireless Access Point
Connects a computer to a network/router
… using Wi-Fi/Bluetooth
… in a hotspot
… using radio transmitters/antennae
(3 × [1])

Switched hub
Acts as a connection point for a number of network computers
Checks the destination/IP address
… of data packets
... and forwards them to the intended recipient
(3 × [1]) [9]

(b) File server [1]
Provides storage space
… for user files
Manages/controls access to these files
(2 × [1])

Printer server [1]
Controls the queue for each printer
Usually provides a print spooler to manage multiple print jobs
(2 × [1])

Web Server [1]
Hosts websites
...and delivers web content/pages
(2 × [1])

Mail/Email Server [1]
Manages the sending
...receiving
...routing of emails
(2 × [1])

Proxy Server [1]
Acts as an intermediary/gateway between clients/servers
...providing anonymity/security/additional performance
(2 × [1])

DNS/Domain Name Server [1]
Translates domain names into IP addresses
...so users to access websites by human-readable names
(2 × [1])

FTP/File Transfer Protocol Server [1]
Manages the transfer of files
...between clients/servers
(2 × [1])

(2 × [3]) [6]

AVAILABLE MARKS

(c) Each node is attached to two adjacent nodes
A token controls the data being transmitted
The node in possession of the token can add a data message
This is passed from node to node
… until it arrives at its destination
Data travels in one direction only to avoid collisions.
(4 × [1]) [4]

(d) **Features of a P2P network**
There is no dedicated/central server
All computers have equal status
Each computer is a supplier of resources and a consumer of resources

Resource sharing
Computers make some of their resources available to other computers
One computer can request a resource from another computer which will then share this resource

Features of CS network
At least one computer acts as the server
This manages file storage and backup, printer management
It provides application software
All the other computers are clients where users log on to

Resource sharing
A user connected to the server has access to network resources such as a dedicated area to store their data, hardware resources, and application software
Irrespective of the client the user logs onto, the network resources available will be the same

Level	Marking criteria	Marks
Band 2	The candidate • Provides a detailed description of each network • Compares how resources are shared on the two networks • Uses the appropriate Digital Technology terminology accurately throughout the response Presentation, spelling, punctuation and grammar are of a high standard.	[5]–[6]
Band 1	The candidate • Provides a detailed description of each network • Describes in detail how resources are shared on one network • Uses some relevant Digital Technology terminology Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[3]–[4]
Band 0	The candidate • Provides a description of one network which lacks detail **or** provides a description of how resources are shared on one network which lacks detail • Makes limited use of Digital Technology terminology Presentation, spelling, punctuation and grammar are such that the intended meaning is not completely clear.	[1]–[2]

[6]

AVAILABLE MARKS

25

2 **(a)** Authentication rules
Error detection rules
Error correction rules
Signalling rules
Communication rules
Data compression rules
Security rules
(4 × [1]) [4]

(b) **(i)** The OSI model consists of **2 sets** and **7 layers**.
(2 × [1]) [2]

(ii) **Presentation**
Interacts with the Application/Session layers
Converts information to the correct video/audio format
Implements encryption
Compresses/decompresses information
(3 × [1])

Data Link
Deals with functions such as error detection
. . . error correction
. . . creating data blocks
. . . synchronising data blocks
It converts each outgoing packet into a series of bits/a series of incoming bits into data packet
Interacts with the Network/Transport layers
(3 × [1]) [6]

AVAILABLE MARKS

(iii) Each layer performs a specific function
The model specifies this function precisely
...but it does not prescribe how this function is implemented
This is left to software/hardware developers
(3 × [1]) [3]

(iv) Network users have greater choice in DT devices
… as compliant devices from different manufacturers will work together
(2 × [1])

If a manufacturer produces a device which complies with the standard
… they know it can be connected to other manufacturers' devices
(2 × [1]) [4]

(v) **TCP/IP**
TCP/IP dictates how information should be packaged and sent and received
The TCP layer handles file transfer between different computers . . . and splits data into packets and allocates an address to each packet
The IP layer is responsible for transferring packets of data from node to node using IP addresses
It verifies the correct delivery of data
(3 × [1])

Bluetooth
Both must be Bluetooth enabled
Each will display the devices within range
. . . which are set to visible
The mobile device will be selected
The device may require pairing/a passcode
A piconet is created
(3 × [1]) [6] 25

3 **(a)** **Entity**
An object about which data is stored
Suitable example, e.g. Course
(2 × [1])

Attribute
A property of an entity
Suitable example, e.g. courseTitle
(2 × [1]) [4]

(b) **(i)** courseTBL(courseID,CourseTitle, degree) [1]
course_moduleTBL(courseID,moduleID,moduleTitle,tutorID,tutorName) [1] [2]

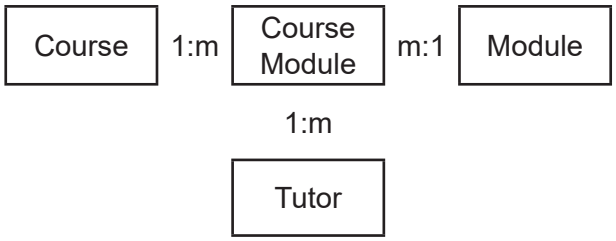
(ii) courseTBL(courseID,CourseTitle, degree) [1]
course_moduleTBL(courseID,moduleID,tutorID,tutorName) [1]
moduleTBL(moduleID,moduleTitle) [1] [3]

(iii) courseTBL(courseID,CourseTitle, degree) [1]
course_moduleTBL(courseID,moduleID,tutorID) [1]
moduleTBL(moduleID,moduleTitle) [1]
tutorTBL(tutorID,tutorName) [1] [4]

(c) Normalisation is a complex/time consuming process
… required to create the database structure
(2 × [1])

A normalised database usually contains more tables than an un-normalised database
… so setting up queries can be more complex/queries and reports may take longer
(2 × [1]) [4]

(d)



[1] for each of 4 entities
[1] for each of 3 relationships
[1] for diagram with no additional components [8]

AVAILABLE MARKS

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4 **(a)** **(i)** **Logical data model**
Produced during the design of the database/is independent of the database software
It identifies the entities
... the relationships between entities
... the attributes for each entity
... the keys for each entity
(2 × [1])

Physical data model
Produced during implementation/defines the physical structure of the database
It creates the table structures
... the column names/field names/aliases
... the data type of each field
... the keys for each table
... validation rules
(2 × [1]) [4]

(ii) Contains descriptions/information about the data held in a database
Attribute/Field name
Data type
Field size
Format
Default value
Entity/table name
Identification of keys
Validation rules
Index(es)
Input masks
(6 × [1]) [6]

AVAILABLE MARKS

(b) (i) firstname varchar (25)
courseCode varchar (4)
(2 × [1]) [2]

(ii) INSERT INTO studentTBL [1]
VALUES (S2222, Knox, Kevin, C303, kknox@emailaddress.com)
[1] + [1]
(3 × [1]) [3]

(iii) UPDATE studentTBL [1]
SET CourseCode = C303 [1] + [1]
WHERE studentID = S9876; [1]
(4 × [1]) [4]

(c) Features of QBE
QBE provides a graphical way of querying a database
The user enters commands, example elements and conditions into a blank template/form to specify fields and values to be used in a query

Features of SQL
SQL is a programming language for creating, accessing and manipulating databases
SQL provides commands to create tables and to insert, search, update, delete records
SQL consists of a large set of commands, each with its own syntax

Evaluation: How queries are created in each
QBE Database users will require minimum technical knowledge/skill
SQL Database users will require a high level of technical knowledge/skill
SQL provides a more comprehensive set of querying tools for more complex queries

Level	Marking criteria	Marks
Band 2	The candidate • Provides a detailed description of each approach which is correct • Evaluates how queries are created in both approaches • Uses the appropriate Digital Technology terminology accurately throughout the response Presentation, spelling, punctuation and grammar are of a high standard.	[5]–[6]
Band 1	The candidate • Provides a detailed description of each approach • Describes how queries are created in one approach • Uses some relevant Digital Technology terminology Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[3]–[4]
Band 0	The candidate • Provides a description of QBE **or** SQL which lacks detail **or** compares superficially how queries are created in both approaches • Makes limited use of Digital Technology terminology Presentation, spelling, punctuation and grammar are such that the intended meaning is not completely clear.	[1]–[2]

[6]

AVAILABLE MARKS

25

5 **(a)** (Human) player asks questions
...of a human and a machine
The player does not know which is the human and which is the machine
If player cannot distinguish between the human and the machine
...the machine can be classified as intelligent
(4 × [1]) [4]

(b) **(i)** **Shell**
The software development environment for creating an expert system
It contains the components of an expert system
... such as a knowledge acquisition system/knowledge base/inference engine/user interface
These components are populated/configured for the particular application
(3 × [1])

Fuzzy logic
Uses probabilities/degrees of truth
... instead of true & false/1 & 0/Boolean logic/formal logic
Decisions can be made with incomplete data/uncertain data
Computers can mimic human reasoning
Fuzzy logic is designed to solve problems by making the best possible decision given the input
(3 × [1])

User interface
Allows a user to communicate with the expert system
The user answers a series of questions relating to the problem/the user can query the system
Possible solutions are presented to the user
… with reasoning/probabilities
(3 × [1]) [9]

(ii) Inference engine
Knowledge base
Inferences/conclusions (either order)
Rules
Subject
(6 × [1]) [6]

(c) Note: Use the terms to explain...
'using a mobile phone to a landline'
Candidates are not required to explain the terms

Stage 1
The caller's phone sends radio signals to the nearest Cell
The cell contains a Mobile Phone Mast and the base station controlled by a Base Station Controller

Stage 2
The radio signals are forwarded to the nearest Mobile Switching Centre
As the caller moves between cells, control moves from one base station to another via handoffs

Stage 3
The call is transferred to the Public switched Telephone Network (PSTN)
...and passed to the appropriate landline

Level	Marking criteria	Marks
Band 2	The candidate • Provides an accurate and clear description of all three stages using all the terms appropriately • Uses the appropriate Digital Technology terminology accurately throughout the response Presentation, spelling, punctuation and grammar are of a high standard.	[5]–[6]
Band 1	The candidate • Provides an accurate and clear description of two the stages using all the relevant terms • Uses some relevant Digital Technology terminology Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[3]–[4]
Band 0	The candidate • Briefly describes one of the stages using some of the relevant terms • Makes limited use of Digital Technology terminology Presentation, spelling, punctuation and grammar are such that the intended meaning is not completely clear.	[1]–[2]

[6]

AVAILABLE MARKS

25

AVAILABLE MARKS

6 (a) (i) **Data mining**
The analysis of large data sets
...to discover patterns/trends
It applies artificial intelligence/statistics/database systems
...to structured and unstructured data sets
...which are difficult to analyse using traditional methods
(2 × [1])

Big data
Complex data sets
...that traditional database applications
...cannot capture/curate/manage/process
...within an acceptable time frame
...because of volume/variety/velocity
...to enable enhanced decision making
(2 × [1]) [4]

(ii) Identifying patterns
...in customer spending
...automatically
Responding to unusual patterns
...in real time
Confirming transactions
...by obtaining confirmation from customer
(3 × [1]) [3]

(b) (i) Use of PINs
...at ATMs/when making purchases
(2 × [1])

Two-factor authentication
...for online banking
(2 × [1])

Biometric access
…when a customer uses a card on a mobile device
(2 × [1])

Use of encryption
…when transmitting data
(2 × [1])

Implement staff training
...to reinforce the need for keeping data secure
(2 × [1])

Implement username and passwords/access levels
...restricting employee access to data
(2 × [1])

(3 × [2]) [6]

(ii) Personal data should be processed fairly and lawfully with the consent of the data subject
Personal data should be used for the specified purpose only
Personal data should be adequate and relevant for its intended purpose
Personal data should be accurate and up to date

		AVAILABLE MARKS
	Personal data should not be kept for longer than necessary Personal data should be processed in accordance with the rights of the data subject Personal data should not be transferred outside the EU (3 × [1]) [3]	
(iii)	The person in the company/organisation ...responsible for controlling the way data is processed ...and informing employees/employer about their responsibilities under the Act (3 × [1]) [3]	
(iv)	Information Commissioner Data subject (2 × [1]) [2]	
(c)	The bank should ensure that employees are aware of the terms of the Act … and the consequences of being in breach of it … via an Acceptable Use Policy … identifying disciplinary procedures (2 × [1]) Computer use by the bank's employees should be audited regularly … and suspect access activity investigated (2 × [1]) Employee access should be restricted using a username and password system ...so that employees only have the access that is necessary for their work (2 × [1]) [4]	25
	Total	**150**